



Study on the characteristics of $W_1/O/W_2$ emulsion delivery system stabilized by soybean protein isolate/(-)-epigallocatechin gallate/exopolysaccharides complex

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ABSTRACT

In this study, a ternary complex was prepared using soybean protein isolate (SPI), (-)-epigallocatechin gallate (EGCG), and exopolysaccharides (EPS) produced by *Lacticplantibacillus plantarum* A114. It systematically evaluated the stabilization on $W_1/O/W_2$ emulsions and protective effects on the encapsulated *L. plantarum* A114 upon long-term storage, pasteurization, and *in vitro* simulated gastrointestinal digestion. The physical stability and microstructure of the emulsions were analyzed by particle size, ζ -potential, creaming index, and rheological behaviors. The survival rate of the strain was determined by flow cytometry during 28-day of storage (4 °C), after pasteurization and *in vitro* simulated gastrointestinal digestion to assess their protective efficacy. The results showed that compared with the emulsions stabilized by native SPI, (EGCG-SPI)-EPS stabilized emulsion was with the smallest particle size and the lowest degree of emulsion separation during 28-day storage period, and the survival rate of *L. plantarum* A114 increased by 1.23 times during storage, 0.88 times after pasteurization, and 1.45 times after *in vitro* simulated gastrointestinal digestion. These results suggested that the emulsions stabilized by (EGCG-SPI)-EPS complex exhibited superior physical stability and protective effects on encapsulated strains. In addition, this study further observed the microstructure of the strains encapsulated in the emulsion stabilized by different complex using confocal laser scanning microscopy (CLSM). The results indicated that the introduction of EGCG and EPS could construct a composite interface layer with stronger physical barriers and anti-oxidant protection through the synergistic effect with SPI, thereby significantly improving the stability of $W_1/O/W_2$ emulsion system and the protective effect on *L. plantarum* A114. To sum up, the (EGCG-SPI)-EPS ternary complex presented the best protective synergistic effect and was an extremely promising targeted delivery system for probiotics. This study has important guiding significance for the development of probiotic functional foods with high survival rates.

1. Introduction

Probiotics referred to active microorganisms which could colonize in intestine and had beneficial effects on human health when consumed in sufficient quantities (Hill et al., 2014). Numerous studies had shown that appropriate supplementation of probiotics not only helped regulate intestinal microecology and inhibit the growth of pathogenic bacteria, but also enhanced the immune barrier function of the body and played a preventive and auxiliary therapeutic roles in various gastrointestinal diseases, thereby comprehensively promoting the health of the host

(Jiang et al., 2022). *Lactobacillus plantarum*, as an important member of probiotics, has been widely used in functional foods and health supplements due to its excellent acid and bile salt resistance and various probiotic functions (such as regulating the intestinal flora and enhancing immunity, etc.) (Dell'Anno et al., 2021). Relevant study had indicated that the probiotics could only exert their probiotic effects when the number of live bacteria released into human intestinal tract reached at least 10^6 CFU/g or more (Flach et al., 2018). However, most probiotics needed to withstand pasteurization during food processing, environmental stress during long-term storage, and extreme conditions such as

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strong stomach acid and bile salts in the digestive tract during their journey to the colon, resulting in a obvious decrease in the number of live bacteria and severely limiting the exertion of their biological activity (Min et al., 2019). Therefore, it had become a key issue in current research and application to explore how to effectively protect probiotics, and increase their survival rate and stability during processing, storage, and in the digestive tract environment.

Among various probiotic protection strategies, encapsulation technology was widely regarded as an effective method (Neuenfeldt et al., 2022). This technology not only significantly enhanced the survival rate of probiotics during processing, storage, and transportation, but also enabled targeted delivery and precise release of probiotics through the design of encapsulation structures or materials, thereby accurately exerting their probiotic effects. Common encapsulation techniques included spray drying, freeze drying, extrusion, and emulsification (Shofinita et al., 2023). However, traditional methods such as spray drying and freeze drying often involved high temperatures or strong mechanical stress, which could cause damage to the bacteria, and high costs. In contrast, emulsion encapsulation technology was more gentle and conducive to maintaining the activity of probiotics (Su et al., 2021). Among various emulsion systems, $W_1/O/W_2$ emulsion, due to its unique “two membranes and three phases” structure, demonstrated significant advantages in encapsulation and controlled release of water-soluble active components. This system encapsulated probiotics within the internal water phase of $W_1/O/W_2$ emulsion, thereby avoiding contact with adverse external factors, while the oil and external water phases provided dual protection for the probiotics. For example, the $W_1/O/W_2$ emulsion encapsulating *L. acidophilus* showed a significant increase in the survival rate of the encapsulated bacteria in an acidic environment, with a survival rate of up to 84.24% after *in vitro* gastrointestinal simulation digestion, fully verifying the protective potential of this system (Wang et al., 2020).

Although the $W_1/O/W_2$ emulsions presented unique advantages in encapsulating probiotics, their practical application was still limited by the inherent physical instability of the system, which often led to flocculation, aggregation, and emulsion breakdown, resulting in structural damage and encapsulation failure. Therefore, improving the stability of this type of emulsions had become a key research direction to promote its development and application. In order to address the aforementioned stability challenges, various strategies, primarily including physical structure reinforcement and interfacial engineering, etc., have been developed to overcome the thermodynamic instability. For instance, one common and effective strategy was to use food-grade colloidal particles (such as gliadin nanoparticles) to stabilize Pickering emulsion, which constructed a mechanical barrier by forming an irreversible adsorption layer at the oil-water interface, thereby inhibiting droplet coalescence (Zhang et al., 2022). Another widely employed strategy to stabilize emulsion was behavior modulation, including gelation of the internal aqueous phase (e.g., using gelatin or gellan gum to form an internal hydrogel) and crystallization of the oil phase (e.g., using high-melting-point lipids to form a solid shell) (Elaine et al., 2024). These approaches restricted the migration of the internal aqueous phase and the ostwald ripening of droplets, significantly improving storage stability (Chen et al., 2018). Furthermore, advanced preparation techniques such as microfluidics enabled precise control over droplet size and structural homogeneity, offering the possibility of producing monodisperse, single-core high internal phase $W_1/O/W_2$ emulsions, thereby enhancing the stability of the system from a structural origin (Guo et al., 2024). However, these strategies could effectively improve the physical stability of emulsions, but their functional design remained largely focused on constructing physical barriers, and the active protective functions they provided for encapsulated active substances, especially those sensitive to oxidation, heat, and gastrointestinal environments, such as probiotics, were still relatively limited.

SPI, as a high-quality plant-based emulsifier, could effectively adsorb onto the oil-water interface to form an interface film. However, its

ability to resist environmental disturbances was poor when used alone, and it was prone to aggregation when exposed to temperature, ionic strength, or pH changes, causing emulsion stratification or demulsification (Nguyen et al., 2017). To overcome this limitation, researchers have made efforts to combine SPI with polyphenols or exopolysaccharides (EPS) to construct a more superior composite interface film (Hu et al., 2021). EGCG, as the most active monomer of tea polyphenols, not only had excellent antioxidant activity but also could bind to proteins through covalent or non-covalent interactions, improving their functional properties and enhancing the mechanical strength of the interface film. However, it was still relatively limited relying solely on EGCG modification to enhance the stability of the emulsion environment and storage stability. Further research had found that introducing EPS into the protein-polyphenol complex could not only inhibit droplet migration through viscosity enhancement but also might undergo electrostatic deposition or synergistic adsorption with interface proteins, forming a multi-layer interface structure with stronger spatial steric hindrance, ultimately achieving a synergistic enhancement of stability. In previous studies, we found that *L. plantarum* A114 EPS had good physical and biological activities, and the identification results of *L. plantarum* A114 had been shown in the supplementary materials of our previous study (Duan et al., 2025). The ternary complex prepared using protein, polyphenols, and EPS had the advantages of multiple components and was endowed with multiple functional attributes. Among them, protein served as the emulsifying basis, polyphenols provided antioxidant properties, and EPS provided spatial stability, all of them jointly maintained the stability of the emulsion while expanding its functional properties. Currently, research had mainly focused on the interaction between SPI and polyphenols or EPS, but there were few studies on preparing ternary complexes by combining the molecular modification enhancement effect of EGCG with the physical barrier-building ability of EPS to enhance the stability of $W_1/O/W_2$ emulsions and the protective effect of probiotics. There was also a lack of in-depth and systematic research on the synergistic enhancement mechanism of the ternary complex for the stability of $W_1/O/W_2$ emulsions and the protective effect of probiotics.

Therefore, in this study, *L. plantarum* A114 was used as the targeted delivery strain. SPI, EGCG-SPI, SPI-EPS, and (EGCG-SPI)-EPS were respectively utilized to stabilize the $W_1/O/W_2$ emulsion, thereby constructing four kinds of emulsion delivery systems. It aimed to systematically evaluate the physical stability (particle size, ζ -potential, creaming index, rheology behaviors) and microscopic structure (CLSM) changes of the emulsions, and combine with the analysis of the impact of the emulsions on the survival rate of probiotics during pasteurization, long-term storage, and *in vitro* simulated gastrointestinal digestion to finally explore the improving mechanism of the emulsion interface properties by different complexes and their protective efficacy on probiotics. This study provides theoretical basis and practical reference for the development of new, efficient, and stable probiotic-targeted delivery systems.

2. Materials and methods

2.1. Materials and reagents

SPI ($\geq 85\%$, 20–100 kDa), pepsin (1:3000) and trypsin (1:250) were bought from Solarbio Life Sciences & Technology Co., Ltd. (Beijing, China). EGCG ($\geq 98\%$) was provided by Yuanye Bio-Technology Co., Ltd. (Shanghai, China). *L. plantarum* A114 was stored in the Key Laboratory of Agricultural Product Processing of Changchun University and EPS was extracted from *L. plantarum* A114. Soybean oil was purchased from the local supermarket. Polyhedral ricinoleic acid glycerol (PGPR) was purchased from Qingyuan Food Additives Co., Ltd. (Beijing, China). The LIVE/Dead™ BacLight™ Bacterial Viability kit was provided by Thermo Fisher Scientific Co., Ltd. (Massachusetts, USA). Nile Red and Nile Blue chloride were purchased from Aladdin Biochemical

Technology Co., Ltd. (Shanghai, China).

2.2. Cultivation of bacteria

MRS broth medium was sterilized under high-pressure at 120 °C for 20 min, then *L. plantarum* A114 (5%, V/V) was incubated anaerobically at 37 °C for 24 h. To harvest bacterial cells, the culture was centrifuged at 4 °C, 5000 ×g for 10 min. The precipitate was resuspended in PBS and adjusted to 1×10^9 CFU/mL, followed by vortexing for 10 min to ensure thorough mixing.

2.3. Extraction of EPS

L. plantarum A114 was firstly passaged for 2–3 generations and then cultivated in an incubator shaker at 250 rpm for 48 h. The bacterial suspension was centrifuged, and the supernatant was collected. Trichloroacetic acid (80%) was added to the supernatant at a final concentration of 4% (V/V), followed by storage at 4 °C for 12 h to remove proteins. Rotary evaporation was then performed to reduce volume. Three volumes of pre-cooled 95% ethanol were added, and the mixture was kept at 4 °C for 12 h. After centrifuging at 7000 ×g, 4 °C for 20 min, the precipitate was collected and dissolved in little deionized water, followed by dialyzing and lyophilization to obtain EPS from *L. plantarum* A114.

2.4. Preparation of covalent complexes

The covalent complex was prepared according to the method reported by Duan et al. (2025). The schematic diagram to prepare complexes was shown in Fig. 1. Firstly, the EGCG-SPI complex was prepared by dissolving 1 g SPI in 50 mL distilled water and adding a certain amount of EGCG. 1 M NaOH was used to adjust pH to 9.0, and the mixture was stirred for 24 h. After terminating the reaction by adjusting the pH to 7.4, unbound EGCG was removed by dialysis (8000–14,000 Da, 4 °C, 48 h).

Secondly, 20 mg/mL SPI or EGCG-SPI complex solution was mixed with 20 mg/mL EPS at 1:1 (w/w) and stirred at 25 °C for 3 h. The pH was adjusted to 7.0 with 1 mol/L HCl or NaOH. Finally, the mixtures were lyophilized and incubated at 60 °C for 24 h under 79% relative humidity to obtain SPI-EPS and (EGCG-SPI)-EPS complexes.

2.5. Structural characterization of covalent complexes

The characterization of the complex structure, including intrinsic fluorescence (the emission spectrum was scanned at an excitation wavelength of 280 nm to analyze the conformational changes of the protein), ultraviolet absorption spectra (scanned within the range of 200–450 nm to observe the changes in the characteristic absorption peaks of different samples), Fourier infrared absorption spectra (scanned within 4000–400 cm^{-1} to analyze the changes in functional groups), ζ -potential (the surface charge changes of the complex were evaluated using a nanoparticle size and potential analyzer), differential thermal analysis (scanned under a nitrogen atmosphere at a certain heating rate to analyze the changes in thermal stability), etc., could be found in our

previous study (Duan et al., 2025).

2.5.1. Particle size

The particle sizes of native SPI and various complexes were determined using a particle size analyzer (Brookhaven, New York, USA) (Dong & Hua, 2016). The measurement conditions were as follows: temperature 25 °C, equilibration time 2 min, and the refractive indices of water and protein were fixed at 1.33 and 1.46 respectively. The results were expressed as the volume average particle size ($D_{4,3}$).

2.5.2. Turbidity

The turbidity values of native SPI and various complexes were determined according to the method described by Zhang et al. (2023). Where, the absorbance values (i.e. turbidity values) were measured at 600 nm. Before each measurement, the corresponding buffer was used as a blank.

2.5.3. Surface hydrophobicity (H_0)

The surface hydrophobicity of native SPI and various complexes was determined by ANS (1-anilino-8-naphthalenesulfonic acid) method as described by Xu et al. (2021). The sample solutions were diluted to five different concentration gradients (0.02%, 0.04%, 0.06%, 0.08%, 0.1%) with phosphate buffered saline (PBS, 0.01 M, pH 7.0). 4 mL diluted sample solution was mixed with 20 μL ANS solution (8.0 mM) and kept in dark for 15 min. The fluorescence intensity of the samples was measured using a fluorescence spectrophotometer (excitation wavelength of 390 nm, emission wavelength of 470 nm, slit width of 5 nm). A graph was plotted with the sample concentration (mg/mL) as the x-axis and the fluorescence intensity value as the y-axis. The slope of the curve was the surface hydrophobicity of the sample.

2.6. Preparation of the $W_1/O/W_2$ emulsion encapsulating *L. plantarum* A114

The $W_1/O/W_2$ emulsion encapsulated with *L. plantarum* A114 was prepared by a two-step emulsification method. Firstly, *L. plantarum* A114 was washed twice with sterile PBS at 4 °C, followed by re-suspending in 0.15 M NaCl solution as the inner aqueous phase. Secondly, the oil phase was prepared by dissolving 5% (v/v) polyglycerol polyglycerol ester in soybean oil at 65 °C under magnetic stirring for 15 min. Subsequently, the inner aqueous phase and the oil phase were mixed in a 2:8 (v/v) ratio. Under an ice bath condition, the mixture was homogenized with a homogenizer at 10,000 rpm for 2 min to obtain the water-in-oil primary emulsion. Then, the above primary emulsion was mixed with SPI and various complex solutions in a 7:3 (v/v) ratio respectively. Under the same ice bath condition, the mixture was homogenized with a homogenizer at 6000 rpm for 1 min to successfully obtain the $W_1/O/W_2$ emulsion encapsulating *L. plantarum* A114.

2.7. Structural characterization of $W_1/O/W_2$ emulsion encapsulating *L. plantarum* A114

2.7.1. Particle size and ζ -potential

The particle size and ζ -potential of the emulsion were measured as

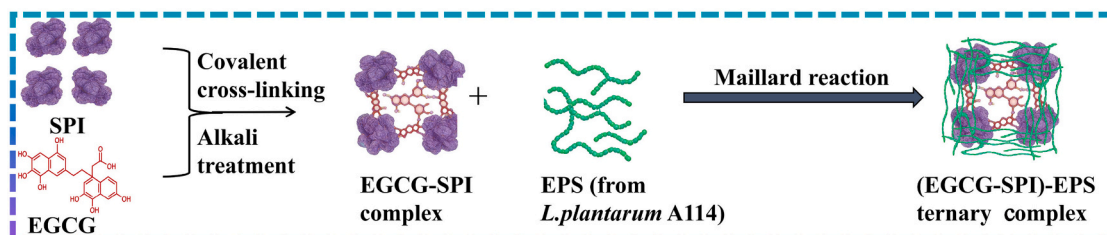


Fig. 1. The schematic diagram to prepare complexes.

description in 2.4.1. To avoid the multiple scattering effect, the sample was diluted to 1:1000 with distilled water before measurement.

2.7.2. Encapsulation efficiency (EE)

The encapsulation efficiency was determined according to the method described by Zhang et al. (2016) with some modifications. Firstly, $W_1/O/W_2$ emulsion was thoroughly vortexed for 5 min to disrupt the emulsion droplets and release the *L. plantarum* A114 in the internal aqueous phase. The broken emulsion was collected and diluted with 0.15 M sterile NaCl in a gradient way, then spread on MRS agar plates and cultured at 37 °C for 48 h. The number of colonies obtained was recorded as the total viable bacterial number (N_{Total}). Another sample (intact emulsion) was diluted with the same volume of sterile saline solution, followed for plate counting. The number of colonies obtained was recorded as the number of free viable bacteria that were not encapsulated (N_{Free}). The EE was calculated using the following formula:

$$EE (\%) = \left(1 - \frac{N_{Free}}{N_{Total}}\right) \times 100\% \quad (1)$$

2.7.3. Microstructure of $W_1/O/W_2$ emulsion

2.7.3.1. Optical microscope. The microstructure (OLYMPUS IX51, Japan) of the $W_1/O/W_2$ emulsion interface was observed using an optical microscope at 20× magnification. Before the observation, the emulsion was diluted at 1:100 with PBS to ensure an appropriate number of droplets in the field, making it easier to observe.

2.7.3.2. Confocal laser scanning microscope (CLSM). CLSM was used to more clearly observe the encapsulation morphology of $W_1/O/W_2$ emulsion to *L. plantarum* A114. Firstly, *L. plantarum* A114 was resuspended in SYTO-9 staining solution, incubated at 37 °C in the dark for 30 min. Then, the excess staining solution was removed by centrifugation at 5000 ×g for 10 min, and the bacteria were resuspended in PBS. Subsequently, this stained bacterial sample was used to prepare $W_1/O/W_2$ emulsion. Then, 10 µL of Nile Red (0.1%, v/v, used to label the oil phase) and 10 µL of Nile Blue chloride (0.2%, v/v, used to label the water phase) were added, followed by incubating at room temperature in the dark for 30 min. 10 µL of the stained emulsion sample was dropped onto a glass bottom cell culture to immediately observe and the microscopic morphology images were collected for further analysis.

2.7.4. Rheological properties

The rheological properties were evaluated as reported by Zare et al. (2025) with slight modification. Specifically, the freshly prepared emulsion was left to stand at 4 °C for 2 h, then measured by a dynamic shear rheometer (TA Instruments-Waters LLC, USA), and all measurements were carried out under a constant temperature of 25 °C. The instrument was equipped with a parallel plate measurement system (plate diameter 40 mm, gap height 1 mm). Take 1.0 mL sample and place it in the center of the lower plate. Firstly, a static flow test was conducted, with a shear rate from 0.1 to 100 s⁻¹, to characterize the viscosity of the sample. Then, a dynamic oscillation test was performed within the linear viscoelastic region (constant strain 0.1%) with a frequency scan at 0.1 to 100 Hz to obtain the storage modulus (G') and loss modulus (G'').

2.8. Determination of oxidation stability of emulsions

The prepared emulsion was placed in dark at 60 °C to promote oxidation reaction. The contents of hydrogen peroxide and malondialdehyde in the samples were measured every 24 h for a total of 7 days.

Determination of primary oxidation products-peroxide value (PV) (Ghohestani et al., 2023): Mix 200 µL emulsion with 1.5 mL mixture of isooctane and isopropanol (3:1, v/v), then vortex for 3 times and centrifuge at 5000 ×g for 5 min. Take 200 µL supernatant (organic

phase) to mix with 2.8 mL mixture of methanol and butanol (2:1, v/v), then add 50 µL ferrous chloride solution (28.6 mg/mL) and 50 µL thiocyanate ammonium solution (0.3 g/mL) to react for 10 min, followed by measuring absorbance at 510 nm. The concentration of peroxide was calculated using the standard curve of hydrogen peroxide isopropylbenzene. The calculation formula for PV was as follows:

$$PV (\text{meq/kg oil}) = \frac{C - C_0}{m \times \frac{V_2}{V_1} \times 55.84 \times 2} \quad (2)$$

where C represented the total concentration of Fe³⁺ produced by hydrogen peroxide in the sample (mg/L), C₀ was the iron concentration of the control sample (mg/L), m was the mass of oil contained in 200 µL emulsion (g), V₁ was the total volume of the organic phase, V₂ was the sampling volume during measurement, 55.84 was the molecular weight of Fe, and 2 was the constant coefficient.

Determination of secondary oxidation products - thiobarbituric acid value (TBA) (Hu & Zhong, 2010): Take 200 µL emulsion to mix with 2.5 mL thio-barbituric acid solution (0.02 mol/L), and 2.5 mL trichloroacetic acid mixture. The mixture was firstly heated in a water bath at 95 °C for 30 min, then cooled in ice water for 10 min. After centrifugation (5000 ×g for 10 min), the absorbance value of the supernatant was measured at 532 nm. The concentration of TBA was calculated using 1,1,3,3-tetraethoxypropane standard curve. The TBA value was calculated according to the following formula:

$$TBA (\text{mg MDA/kg oil}) = \frac{c \times V \times D}{m} \quad (3)$$

where c represented the concentration of malondialdehyde in the sample obtained from the standard curve (mg/mL), V was the total volume of the colour development reaction system, D was the dilution factor, and m was the mass of the emulsion sample (g).

2.9. Determination of storage stability of emulsions

2.9.1. Creaming index (CI)

To investigate the long-term stability of the $W_1/O/W_2$ emulsion at 4 °C, the freshly prepared emulsion was aliquoted into 10 mL sterile glass tubes and stored at 4 °C for 28 days. During this period, the height of emulsion precipitation (H_t) was measured on the 1st, 7th, 14th, and 28th days, and compared with the initial height (H₀). CI was calculated based on the formula:

$$CI (\%) = \frac{H_0 - H_t}{H_0} \times 100\% \quad (4)$$

2.9.2. Particle size

To assess the physical stability during storage, the particle size of the emulsion was measured on days 1, 7, 14, and 28 following the method described in 2.6.1.

2.9.3. The survival rate of strain encapsulated in emulsion

The storage stability of $W_1/O/W_2$ emulsion at 4 °C for 28-day was measured. The changes of the viable count during storage were detected using a flow cytometer on days 1, 7, 14, and 28. *L. plantarum* A114 without encapsulation was used as a control (Su et al., 2021). Firstly, 1 mL emulsion sample was thoroughly vortexed for 5 min to disrupt the emulsion structure (Wang et al., 2020). Then, it was centrifuged at 4 °C and 5000 ×g for 10 min to collect bacterial precipitate. The supernatant was discarded, and the bacterial precipitant was resuspended in PBS and appropriately diluted to ensure that the bacterial concentration for flow cytometer detection was within the ideal range. Subsequently, the LIVE/Dead™ BacLight™ Bacterial Viability kit was used for staining. SYTO-9 and propidium iodide (PI) were added to the bacterial suspension at the recommended ratio of the operating manual, and the mixture was incubated at 37 °C in dark for 15 min. SYTO-9 could label all bacteria

(both live and dead), while PI could only penetrate damaged membrane structures of dead bacteria and quench the fluorescence of SYTO-9 through a fluorescence competition mechanism, thereby achieving the effective distinction between live and dead bacteria. After staining, the samples were detected by a flow cytometer, and fluorescence signals were collected in FTIC (SYTO-9, green fluorescence) and PE (PI, red fluorescence) channels using a 488 nm argon ion laser.

2.10. Determination of pasteurization stability of emulsions

The stability of the sample during pasteurization was evaluated to simulate the heat treatment conditions in food processing (Leite et al., 2023). Firstly, the sample was placed in a water bath with constant temperature and heated until the center temperature reached 63 °C, keeping for 30 min to complete the pasteurization process. Then the sample was immediately transferred to a cold water bath for rapid cooling to terminate the thermal effect. Bacterial counts were conducted according to the method described in Section 2.8.3.

2.11. Stability under in vitro simulated gastrointestinal digestion conditions

The *in vitro* simulated gastrointestinal digestion was detected as proposed by Zong et al. (2023). *L. plantarum* A114 suspended in sterile NaCl solution was used as the control. Firstly, simulated gastric digestion: 1 mL emulsion sample or bacterial solution was mixed with 4 mL simulated gastric fluid (containing 0.3% pepsin and 0.85% NaCl, pH 2.0), and incubated at 37 °C in a constant temperature shaker for 2 h. Secondly, simulated intestinal digestion: 1 mL sample after simulated gastric digestion was mixed with 4 mL simulated intestinal digestion (containing 1% trypsin, 0.68% KH₂PO₄ and 0.03% bile salts, pH 7.0), and incubated at 37 °C for another 2 h (Luan et al., 2023).

To accurately assess the survival rate of *L. plantarum* A114 after digestion by *in vitro* simulated gastrointestinal digestion, live bacteria counting was performed using a flow cytometer. The positive control was prepared by boiling the live bacteria for 10 min, and the bacteria that were not boiled or digested by gastrointestinal juice served as the control group. Firstly, after completing the *in vitro* simulated gastrointestinal digestion, 1 mL of digestion sample was immediately collected and centrifuged at 4 °C, 5000g for 10 min to collect the bacterial strain, followed by diluting to an appropriate concentration by PBS solution for flow cytometer analysis. Subsequently, the changes in the viable count of *L. plantarum* A114 after *in vitro* simulated gastrointestinal digestion were studied according to the method described in Section 2.8.3.

2.12. Statistical analysis

All the experimental data were measured at least three times, and the results were expressed as the mean $\pm$ standard error (SEM). Differences among groups were assessed by one-way analysis of variance (ANOVA) using SPSS 26.0. $p < 0.05$ was defined as statistically significant.

3. Results and discussion

3.1. Particle size analysis

The particle size detection results of native SPI and various complexes were shown in Fig. 2A. The average particle size of native SPI was 259.33 ± 3.30 nm. When SPI was combined with EGCG, its particle size increased significantly. This was because multiple phenolic hydroxyl groups of EGCG undergone an oxidative cross-linking reaction with the amino acid residues of SPI (such as lysine, arginine, and cysteine) through alkaline treatment. One EGCG molecule could act as a “bridge” to connect multiple protein molecules, or bind to multiple sites of a protein molecule. Additionally, the interaction between EGCG and proteins might cause the tight structure of SPI to become loose, resulting in a larger particle size (Jöbstl et al., 2004). After adding EPS, the particle size of SPI-EPS complex was significantly larger than that of SPI and EGCG-SPI complexes, which contributed that EPS formed polysaccharide chains on the surface of SPI through Maillard reaction, and these grafted polysaccharide chains fully extended in the solution to form a hydration layer and generated a huge spatial obstruction, finally greatly increasing the hydration size of the complex and leading to a significant increase in particle size (Ke & Li, 2023). Furthermore, the particle size of (EGCG-SPI)-EPS complex significantly increased due to the internal cross-linking caused by EGCG and the external extension effect caused by EPS, which led to a significant increase in the final particle size of complex (Li et al., 2024).

3.2. Turbidity analysis

Turbidity represented a solution's ability to scatter light. It mainly depended on the concentration, size, shape of the scattering particles, as well as the difference in refractive index between the particles and the solvent. Fig. 2B showed the changes in turbidity of native SPI and various complexes. In this study, the turbidity of SPI was 0.19 ± 0.02 , but significantly raised for EGCG-SPI complex. This was mainly due to the dehydrogenation of the hydroxyl, amino, and mercapto groups of SPI, which then connected with the hydroxyl groups of EGCG, forming more insoluble complexes. These large particles caused strong scattering of light, thereby leading to an increase in turbidity (Xu et al., 2023). Meanwhile, SPI and EPS covalently bound through Maillard reaction to

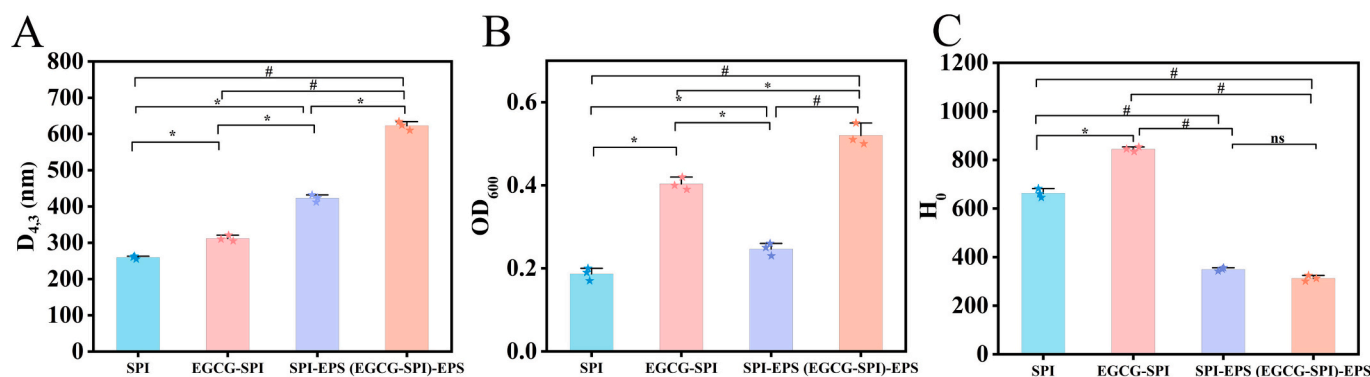


Fig. 2. Particle size (A), Turbidity (B) and Surface hydrophobicity (H₀) (C) of native soybean protein isolate (SPI), soybean protein isolate-(–)-epigallocatechin gallate (EGCG-SPI), soybean protein isolate-exopolysaccharides (SPI-EPS), and soybean protein isolate-(–)-epigallocatechin gallate-exopolysaccharides ((EGCG-SPI)-EPS) complexes. * indicates significant differences at $p < 0.05$, and # indicates significant differences at $p < 0.01$.

form larger complexes with higher molecular weights, leading to an increase in turbidity (Feng et al., 2024). Moreover, it formed a more soluble and more stable colloidal system rather than dense and highly refractive difference aggregates, therefore this increase was uniform and stable. Interestingly, the turbidity of the (EGCG-SPI)-EPS complex was the highest. It was speculated that EPS provided certain spatial stabilizing effect, but it might not be sufficient to completely disassemble or inhibit the formation of huge and dense aggregate structures by EGCG. This huge molecular structure, accompanied with an uneven internal refractive index, formed a very strong light scattering center, finally resulting in a significant increase in turbidity.

3.3. Surface hydrophobicity (H_0) analysis

Surface hydrophobicity could affect the intermolecular forces and reflect the conformation and functional properties of protein (De la Cruz Torres et al., 2022). In this study, the hydrophobicity of native SPI and various complexes was determined using the ANS fluorescence probe method. As shown in Fig. 2C, the surface hydrophobicity of the EGCG-SPI complex was significantly higher than that of SPI, indicating that EGCG induced the unfolding of protein structure to expose more hydrophobic sites, thereby resulting in a significant increase in surface hydrophobicity (Chen et al., 2019). Additionally, the surface hydrophobicity of the SPI-EPS complex was significantly lower than that of SPI, which was due to the physical coverage and shielding of the hydrophobic sites of SPI by EPS through the Maillard reaction, reducing the number of hydrophobic sites in the ANS-bound hydrophobic groups and finally resulting in a decrease of surface hydrophobicity. Comparing to SPI and EGCG-SPI, it was found that the surface hydrophobicity of the (EGCG-SPI)-EPS covalent complex was lower, possibly because most of the hydrophobic sites exposed by EGCG were successfully masked by EPS (Wu et al., 2022). Therefore, the surface hydrophobicity of (EGCG-SPI)-EPS fell to a lower level, but it was comparable to SPI-EPS. As for (EGCG-SPI)-EPS complex, although the addition of EGCG caused a dramatic hydrophobicization, the surface properties of the (EGCG-SPI)-EPS complex was ultimately lied on the hydrophilic modification effect of EPS. Interestingly, the surface hydrophobicity of (EGCG-SPI)-EPS complex was decreased, while the turbidity was the highest. This indicated that the encapsulation by EPS only changed its surface chemical properties (lowering the surface hydrophobicity), but could not change its huge physical size and uneven internal structure.

3.4. Structural characteristic of $W_1/O/W_2$ emulsion

3.4.1. Particle size and ζ -potential analysis

The stability of emulsions was usually related to the particle size. Generally speaking, the smaller the particle size, the more stable the emulsion. As shown in Fig. 3A, the droplet particle size of SPI-stabilized

emulsion was significantly larger than that of the emulsions stabilized by other complexes. Generally, the flocculation of SPI-stabilized emulsion resulted from inadequate electrostatic repulsion to counteract the attractive interactions among droplets. The particle size of the emulsion stabilized by EGCG-SPI complex was decreased. On the one hand, after EGCG and SPI were covalently bonded, the amphiphilicity of the protein was improved, which enhanced its interfacial activity and enabled the formation of smaller initial droplets during homogenization. On the other hand, the negative charges coming from EGCG increased the electrostatic repulsion between droplets which effectively inhibited the flocculation of these small droplets, thereby jointly leading to a reduction in the average particle size of the emulsion. Compared with the control group, the average particle size of SPI-EPS complex-stabilized emulsion was significantly reduced. It was speculated that the EPS in emulsion droplets provided strong spatial repulsion for proteins, thereby reducing the aggregation of droplets (Liu et al., 2016). Furthermore, the average particle size of the emulsion stabilized by (EGCG-SPI)-EPS ternary complex was significantly smaller than that of other groups ($P < 0.05$), indicating that the ternary complex was the most effective emulsifier. Presumably, (EGCG-SPI)-EPS complex combined the surface charge of the polyphenol and EPS provided space repulsion. These results indicated that EGCG and EPS modification improved the emulsion stability of SPI. In fact, these findings were consistent with the results reported by other researcher (Wang et al., 2015).

Moreover, the stability of emulsion was also related to its ζ -potential value. As shown in Fig. 3B, the potential values of all samples were negative. Specifically, the potential of the SPI-stabilized emulsion was -18.09 ± 2.16 mV, and its absolute value was significantly lower than that of the emulsions stabilized by other complexes. Moreover, the (EGCG-SPI)-EPS complex-stabilized emulsion had the largest absolute value, indicating the best stability. The reasons mainly lied in two aspects. Firstly, the phenolic hydroxyl groups on EGCG could partially ionize under alkaline conditions to contribute additional negative charges and simultaneously consume some positive charge sites on the surface of SPI. Secondly, after covalent complexation through Maillard reaction, EPS could effectively mask or weaken any tiny non-uniform charge regions on the surface of oil droplets. Therefore, (EGCG-SPI)-EPS complex carried a large amount of negative charges which increased the electrostatic repulsion within the emulsion system and hindered the aggregation and collision of droplets, thereby enhancing the stability of the emulsion (Li et al., 2020).

3.4.2. Encapsulation efficiency (EE) analysis

Fig. 3C showed the effects of native SPI and various complexes on the EE of *L. plantarum* A114. The EE was only $74.04 \pm 1.42\%$ for SPI-stabilized emulsion, which was significantly lower than that of other groups. This was mainly because that the SPI-stabilized interface film was relatively loose and lacked mechanical strength. At the same time,

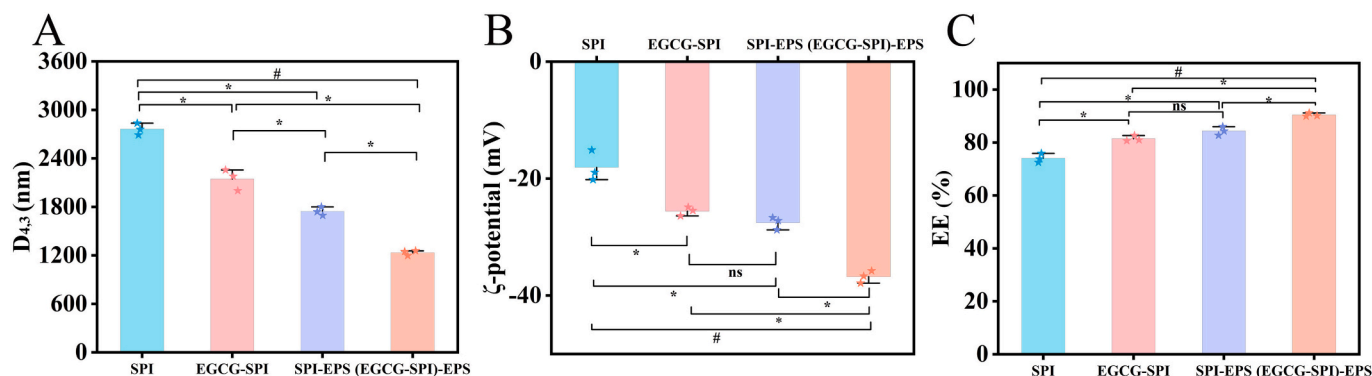


Fig. 3. Particle size (A), ζ -potential (B) and encapsulation efficiency (EE) (C) of the $W_1/O/W_2$ emulsion encapsulating *L. plantarum* A114 was stabilized by native SPI, EGCG-SPI, SPI-EPS, and (EGCG-SPI)-EPS complexes. * indicates significant differences at $p < 0.05$, and # indicates significant differences at $p < 0.01$.

the weak interface film was also unable to effectively prevent the migration of the bacteria between emulsion droplets, resulting in an unsatisfactory *EE*. Among them, the *EE* reached $90.41 \pm 0.55\%$ for the emulsion stabilized by (EGCG-SPI)-EPS complex, reaching the highest value. The increase in *EE* might be due to the increased viscosity and thicker wall layer of the (EGCG-SPI)-EPS-stabilized emulsion provided better protection for *L. plantarum* A114. Relevant study employed ultrasonically treated pea protein combined with pectin for emulsification, relying on high-intensity ultrasound to improve protein dispersion and interfacial adsorption, while pectin increased the system viscosity. The highest encapsulation efficiency reached 59.94%, but the overall protective performance, particularly in terms of physical stability and synergistic protection against multiple stresses, was inferior to that of the ternary complex system (Zhang, Wang, et al., 2024; Zhang, Zhang, et al., 2024). Additionally, the higher *EE* was consistent with the smaller emulsion particle size of the (EGCG-SPI)-EPS group, highlighting the importance of the stable emulsion structure for the *EE* of probiotics (Frakolaki et al., 2020).

3.4.3. Microscopic structure analysis of $W_1/O/W_2$ emulsion

The microscopic morphology and structure of the $W_1/O/W_2$ emulsion encapsulating *L. plantarum* A114 were observed using microscope. As shown in Fig. 4, the “two membranes and three phases” structure could be clearly observed under microscope. At the same time, the

spherical droplet size of (EGCG-SPI)-EPS stabilized emulsion decreased and tended to stabilization, indicating that the complex had a significant influence on the droplet size of the emulsion. It was because that more complex particles adhered to the interface layer to stabilize smaller droplets, and the EPS in the complex also made the entire emulsion system more viscous, further preventing droplet aggregation (Baraki et al., 2021). The CLSM images (Fig. 5) showed that the emulsion sizes of (EGCG-SPI)-EPS group were highly uniform, presenting perfect spherical shapes and were independent each other. The red fluorescence rings at the edges of the droplets were clearer and more continuous, indicating that the interface films were more complete and dense. The majority of the bacterial fluorescence signals were confined within the droplets, and only sporadic bacteria could be seen in the external aqueous phase. *L. plantarum* A114 and the oil phase were respectively stained blue and red, indicating that *L. plantarum* A114 had been successfully encapsulated in the $W_1/O/W_2$ emulsion. Relevant study reported that similar phenomenon could also be observed, where *L. plantarum* 23-1 and the oil phase were respectively stained blue and red, and the *L. plantarum* 23-1 was surrounded by the W/O emulsion (Xu et al., 2024).

3.4.4. Rheological properties analysis

The apparent viscosity and viscoelasticity of the emulsions stabilized by native SPI and various complexes were studied through rheological properties analysis. As shown in Fig. 6A, the apparent viscosity of $W_1/O/W_2$

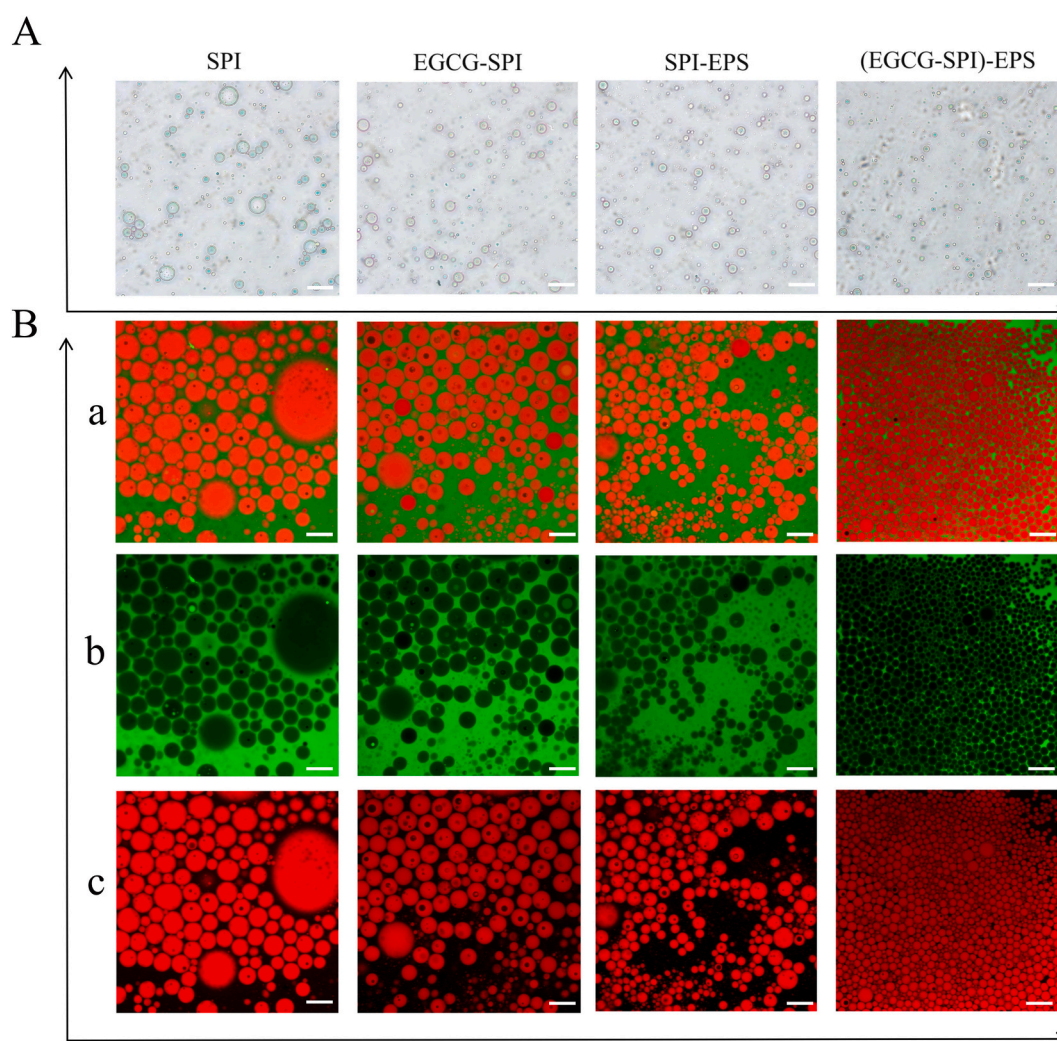


Fig. 4. Optical microscope (A: Scale bar = 50 μ m) and CLSM images (B: Scale bar = 20 μ m) of the $W_1/O/W_2$ emulsion stabilized by native SPI, EGCG-SPI, SPI-EPS, and (EGCG-SPI)-EPS complexes. Where a: The merged images of b and c; b: Water phase stained by Nile Blue, c: Oil phase stained by Nile Red. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

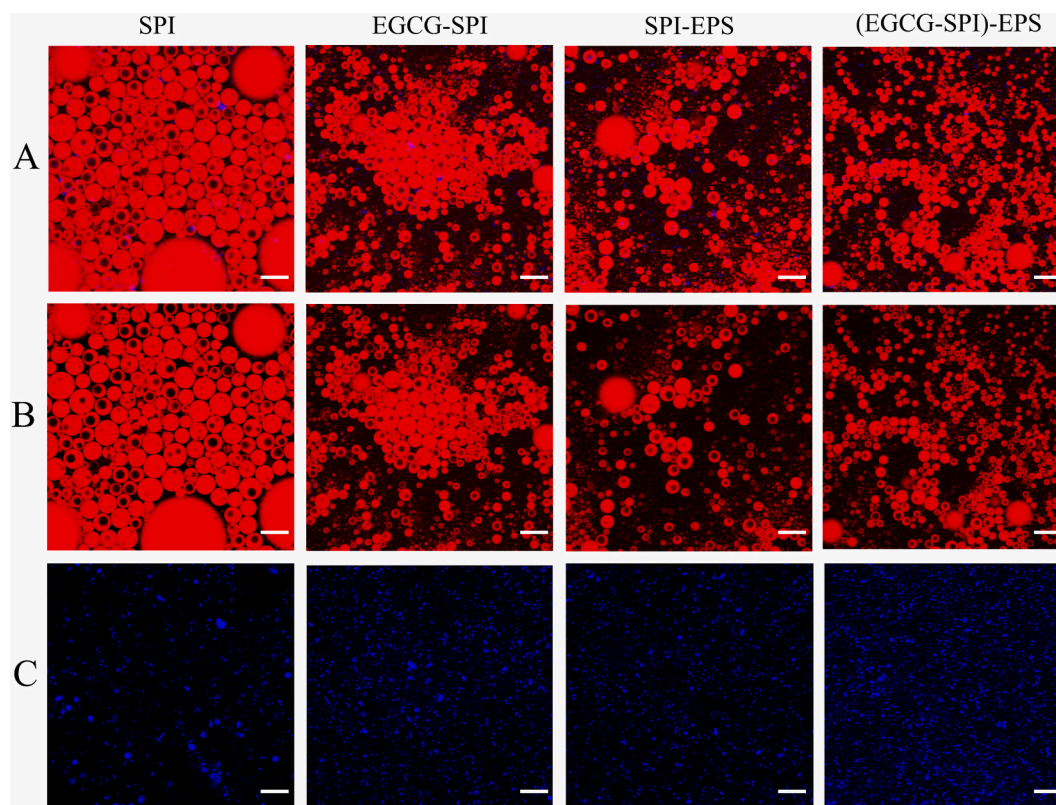


Fig. 5. CLSM images of the $W_1/O/W_2$ emulsion encapsulating *L. plantarum* A114 by native SPI, EGCG-SPI, SPI-EPS, and (EGCG-SPI)-EPS complexes. Scale bar = 20 μm . A: The merged images by B and C. B: Oil phase stained by Nile Red. C: The strains stained with SYTO 9. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

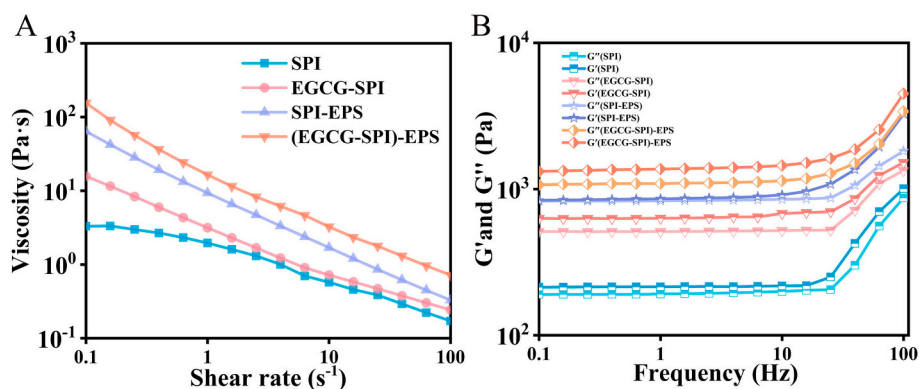


Fig. 6. Viscosity (A), G' (elastic/storage modulus) and G'' (viscous/dissipation modulus) (B) of the emulsions stabilized by native SPI, EGCG-SPI, SPI-EPS, and (EGCG-SPI)-EPS complexes.

O/W_2 emulsion was decreased with the increase of the shear rate, showing a negative correlation. This indicated that the $W_1/O/W_2$ emulsions stabilized by native SPI and various complexes exhibited shear-thinning behaviors, presenting obvious pseudoplastic fluid characteristic. It was speculated that the oil droplets rearranged in a certain direction under the shear force, reducing the influence of flow resistance on oil droplets (Niu et al., 2016). Moreover, upon the destruction of the intact complex network structure in the emulsion by the high shear rate, the reorganization speed of droplets was slower than the rupture speed, making it difficult for small droplets to re-aggregate, finally resulting in a decrease in viscosity (Zhang, Wang, et al., 2024; Zhang, Zhang, et al., 2024). The influences of different complexes on the viscoelasticity of the emulsions were shown in Fig. 6B, the G' (elastic/storage modulus) of all emulsions was greater than G'' (viscous/dissipation modulus) with

increased frequency, indicating that the emulsion system was dominated by elastic response and exhibited typical weak gel-like solid behavior (Li et al., 2023). In addition, the (EGCG-SPI)-EPS stabilized emulsion showed the highest G' and G'' value, implying that this emulsion had the strongest weak gel structure. In contrast, another study which used whey protein isolate-konjac glucomannan non-covalent complexes to stabilized emulsion presented that it relied on the thickening and steric hindrance effects of polysaccharides to improve stability, while the mechanical strength and strain strength of the formed network were relatively weak (Guo et al., 2025).

3.5. Oxidation stability analysis of emulsions

PV was used to measure the accumulation of primary oxidation

products (hydroperoxides), while TBA reflected the content of secondary oxidation products (malondialdehyde) and was directly related to the off-flavor of rancidity. The PV and TBA values of different emulsions stabilized by native SPI or various complexes during storage were presented in Fig. 7A-B. Overall, the PV and TBA values of all emulsions showed an upward trend as storage time increased. Among all groups, SPI stabilized emulsion had the highest PV and TBA values, indicating that the oil was most prone to oxidation, hydrogen peroxides was easy to accumulate rapidly, so that a large amount of secondary oxidation products were generated, resulting in a significant rancid odor and a severe decline in oil quality. This was because the physical barrier effect of SPI interface film was weak and unable to effectively block oxygen, and SPI itself did not have strong antioxidant activity. Compared with SPI-stabilized emulsion, both PV and TBA values of EGCG-SPI stabilized emulsion were significantly decreased, clearly demonstrating the excellent antioxidant ability of EGCG. Meanwhile, the PV and TBA values of SPI-EPS stabilized emulsion were also lower than that of native SPI, but the protective effect was slightly inferior to the EGCG-SPI group. Additionally, (EGCG-SPI)-EPS stabilized emulsion showed the best antioxidant effect, where PV value was significantly lower than all other groups and with a slower increase rate, reflecting the perfect synergy of physical barriers and chemical antioxidants. As for (EGCG-SPI)-EPS stabilized emulsion, EGCG effectively prevented lipid oxidation due to the contribution of hydrogen atom in aromatic hydroxyl group, and the addition of EPS made the emulsion viscous and inhibited the aggregation of oil droplets in the emulsion. Therefore, the (EGCG-SPI)-EPS complex provided physical and chemical barrier for emulsion, so that oxygen could not fully contact and react with the oil, thus delaying oxidation (Li et al., 2023).

3.6. Storage stability analysis of emulsions

3.6.1. Creaming index (CI) analysis

The emulsification degree of the emulsions that encapsulated *L. plantarum* A114 (stabilized by native SPI or other complexes) during storage were shown in Fig. 8A-B. The physical stability of SPI-stabilized emulsion was the poorest, accompanied by rapid and severe phase separation. During the first 7 days, the stability of EGCG-SPI-stabilized emulsion was significantly better than that of SPI-stabilized emulsion, but obvious phase separation could be observed for 7–28 days of storage. This was because the cross-linking of EGCG enhanced the rigidity and strength of the interface film, thus slowing down the flocculation rate of emulsion droplet and showing better stability in the initial stage. However, this interface film still lacked sufficient space hindrance to completely inhibit the slow aggregation of emulsion droplets. Therefore, the droplet size gradually increased as the storage time increased, ultimately leading to obvious emulsion separation. Furthermore, SPI-EPS-stabilized emulsion demonstrated superior long-term stability, with a

very slow phase separation process, and a small amount of phase separation began to appear around 7 days of storage. Its better stability was mainly attributed to the thick and solid condensation layer structure formed by SPI and EPS at the interface, making it difficult for droplets to approach each other due to the repulsion of polymer chains, effectively inhibiting flocculation (Khalloufi et al., 2008). Throughout the storage period, until the 28th day, the CI value of (EGCG-SPI)-EPS-stabilized emulsion was $92.5 \pm 0.55\%$, and only extremely minor phase separation was observed at the bottom of the bottle. The reason lied in that the interface film formed by (EGCG-SPI)-EPS was extremely strong and dense, almost preventing droplet flocculation and size growth, finally eliminating the main cause of phase separation from the root.

3.6.2. Particle size analysis

In Fig. 8C, as for SPI-stabilized emulsion, the initial particle size was the largest and underwent the most significant changes within 28 days, implying the poorest physical stability. This indicated that the mechanical strength of the interface film formed by native SPI was insufficient and unable to effectively resist the van der Waals force between droplets during long-term storage, resulting in severe flocculation and aggregation. Additionally, the rearrangement and desorption of protein molecules might also lead to the instability of the interface film, ultimately causing droplet merging and a significant increase in particle size. The EGCG-SPI complex-stabilized emulsion exhibited significantly good stability than that of SPI-stabilized emulsion, with a faster growth in particle size in the first 7 days and a more gradual increase thereafter. This was because the cross-linking of EGCG and SPI formed a more solid and dense interface film, enhancing the spatial hindrance against aggregation. However, despite the improvement in physical stability, a single protein-based interface still presented slow molecular rearrangement over a long-term storage period. Moreover, there was insufficient electrostatic repulsion or spatial hindrance to completely inhibit droplet proximity, thus still resulting in moderate flocculation. In addition, SPI-EPS complex-stabilized emulsion exhibited better long-term stability, with a slow and minor increase in particle size throughout the storage period. This was mainly due to the formation of a classic composite condensation layer at the oil-water interface between SPI and EPS, providing strong spatial hindrance and effectively preventing the close contact and aggregation of droplets (Li et al., 2021). Interestingly, (EGCG-SPI)-EPS complex-stabilized emulsion displayed the best physical stability, with the particle size remaining almost constant throughout the storage period. This synergistic effect produced extremely strong mechanical strength and spatial hindrance, almost completely inhibiting droplet flocculation and aggregation, ensuring the integrity of the emulsion throughout the storage period. This was consistent with the unnoticeable stratification and stability observed in Fig. 8A.

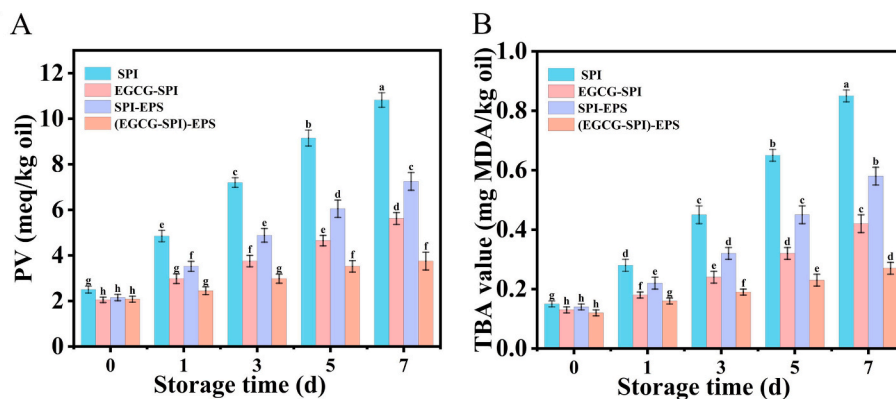


Fig. 7. PV (A) and TBA (B) values of different emulsions stabilized by native SPI, EGCG-SPI, SPI-EPS, and (EGCG-SPI)-EPS complexes. Different letters represent significant differences at $p < 0.05$.

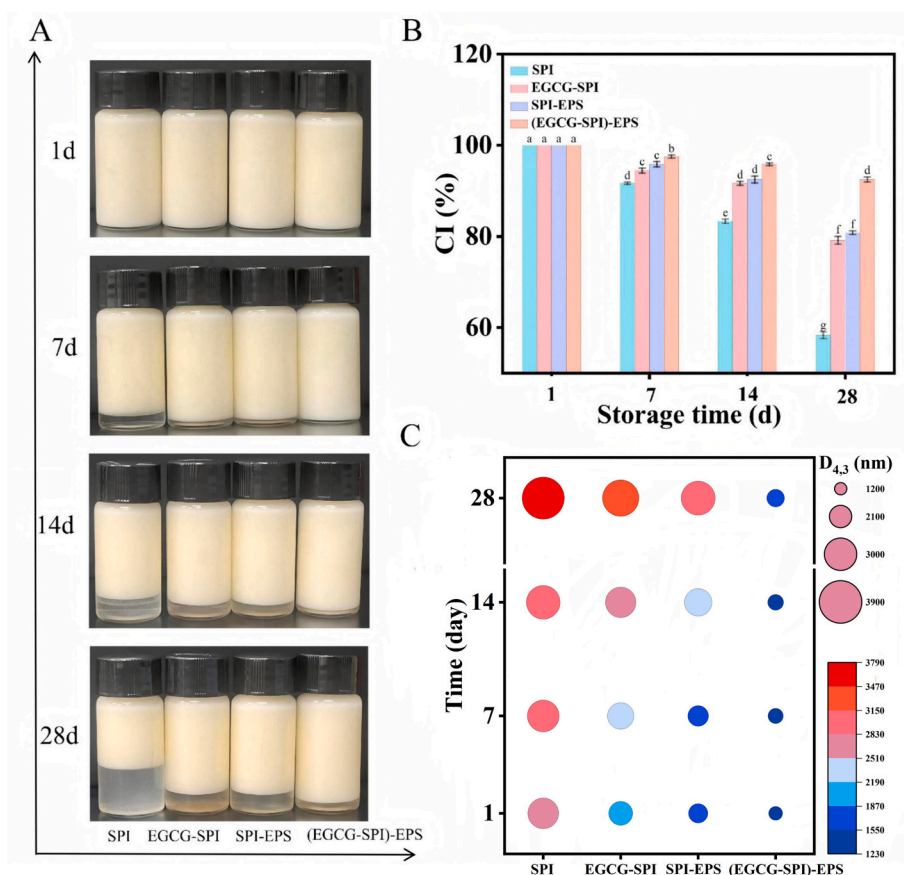


Fig. 8. Creaming index (CI) (A-B) and Particle size (C) of the emulsions stabilized by native SPI, EGCG-SPI, SPI-EPS, and (EGCG-SPI)-EPS complexes were measured on the 1st, 7th, 14th, and 28th days. Different letters represent significant differences at $p < 0.05$.

3.6.3. Analysis of the survival rate of strain encapsulated in emulsion

As shown in Fig. 9, the control group exhibited the fastest decline rate during the 28-day storage period, with a survival rate of only $4.18 \pm 0.19\%$ (Fig. 10A). The SPI-stabilized emulsion provided some protection for encapsulated *L. plantarum* A114, where the survival rate was higher than that of the control group on the 28th day, but the survival rate was still low ($22.0 \pm 0.82\%$), suggesting that SPI interface film could only provide physical protection for a short period, but its structure might be unstable under long-term storage and unable to effectively block the exchange between the internal aqueous phase (W_1) and the external aqueous phase (W_2), resulting in limited protection for the stain. The EGCG-SPI group showed good stability throughout the storage period, with a survival rate of $34.07 \pm 0.74\%$ on the 28th day. It was speculated that EGCG continuously exerted its strong antioxidant activity during storage to effectively eliminate the slowly generated free radicals and protect the integrity of the bacterial cell membrane, thereby significantly slowing down the bacterial death rate. Additionally, the cross-linking of EGCG and SPI made the interface film structure more dense to better maintain the integrity of the emulsion structure under long-term storage, slowing down the leakage and inactivation of the stain in the internal aqueous phase (W_1). Moreover, the SPI-EPS group had a better protective effect than that of EGCG-SPI group, with a survival rate of $40.06 \pm 0.73\%$ on the 28th day. This indicated that the composite condensation layer structure formed by SPI and EPS was very stable and dense, effectively inhibiting the migration of water, the penetration of oxygen, and the invasion of acidic substances, thus creating a relatively independent and stable microenvironment. More importantly, the (EGCG-SPI)-EPS group was with a survival rate of $49.07 \pm 0.82\%$, significantly higher than all other groups ($p < 0.05$), suggesting that (EGCG-SPI)-EPS complex stabilized $W_1/O/W_2$ emulsion

could effectively isolate *L. plantarum* A114 from the adverse external environment, maximizing the survival rate of *L. plantarum* A114. In contrast, a study investigated the whey protein concentrate and pectin electrostatic complex system and found that it primarily relied on a protein-polysaccharide network to enhance interfacial thickness and continuous phase viscosity, functioning mainly as a physical barrier. The post-storage viable bacterial count in their stabilized emulsion was lower than that achieved in our study after conversion (Liang et al., 2023). Hua et al. also found that $W_1/O/W_2$ emulsions stabilized by egg white protein fibrils and pectin significantly increased the survival rate of encapsulated *L. plantarum* during storage (Hua et al., 2024).

3.7. Stability analysis of pasteurization

Pasteurization had a significant impact on the survival rate of *L. plantarum* A114 encapsulated in emulsions stabilized by native SPI or different complexes. As shown in Fig. 10B-C, the majority of stains in control group was inactivated after pasteurization, with a very low survival rate of only $1.26 \pm 0.22\%$. This indicated that *L. plantarum* A114 itself was highly sensitive to heat and pasteurization was sufficient to cause a significant loss of viability without the protection of the emulsion (Zhang et al., 2016). Compared to the control group, SPI-stabilized emulsion provided significant protection, with a survival rate of $11.43 \pm 0.45\%$, demonstrating that a single SPI interface film partially isolated the stains in the internal aqueous phase (W_1) and prevented external heat to some extent. The combination of EGCG and SPI significantly enhanced the protective effect, with a survival rate of $14.09 \pm 0.38\%$, significantly higher than that of SPI group ($p < 0.05$). Presumably, EGCG-SPI complex formed a more dense and mechanically stronger interface film through covalent cross-linking, providing

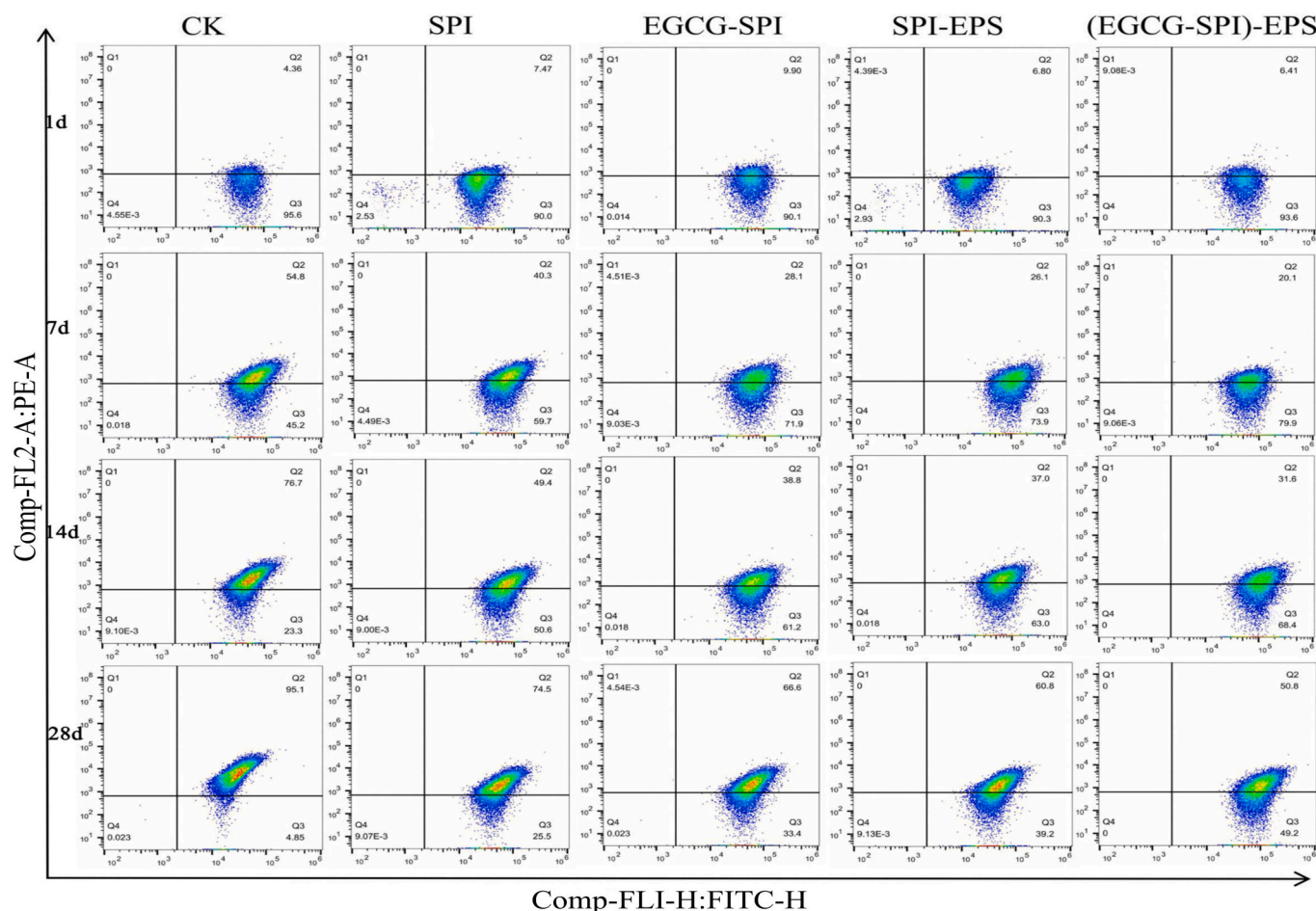


Fig. 9. The survival rates of *L. plantarum* A114 encapsulated in emulsions stabilized by native SPI, EGCG-SPI, SPI-EPS, and (EGCG-SPI)-EPS complexes were detected by flow cytometer on the 1st, 7th, 14th, and 28th days.

superior physical isolation. In addition, as a potent antioxidant, EGCG removed the reactive oxygen free radicals generated during heating, thereby reducing the damage to the bacterial cell membrane caused by oxidative stress. The protective effect of SPI-EPS was comparable to EGCG-SPI ($p > 0.05$), with a survival rate of $14.60 \pm 0.67\%$. Its protective ability mainly stemmed from the formation of a thicker, more stable, and thermally irreversible composite condensation layer at the oil-water interface by the electrostatic interaction between SPI and EPS, which significantly delayed the penetration of heat and increased the viscosity of the external aqueous phase, further hindering heat conduction and providing a better physical barrier (Hua et al., 2023). More importantly, (EGCG-SPI)-EPS complex exhibited the most excellent protective ability, with a survival rate of $21.54 \pm 0.77\%$, significantly higher than that of all groups ($p < 0.05$), clearly demonstrating the synergistic effect of physical barrier enhancement and chemical antioxidant protection. Where EGCG-SPI provided an enhanced primary barrier and internal antioxidant environment, and EPS formed an extremely stable and dense composite film. This multiple protective mechanism minimized the damage of heat and harmful substances to *L. plantarum* A114, achieving better protection. A related study employed S/O/W emulsions stabilized with whey protein isolate/pectin multi-layers and reinforced by calcium cross-linking, yet the survival rate of *L. salivarius* NRRL B-30514 was significantly lower than that in this study, indicating relatively weaker thermal protection (Zhang et al., 2015). In summary, the $W_1/O/W_2$ emulsion based on the ternary complex demonstrated superior protective performance during pasteurization through the synergistic effect of physical barrier and chemical antioxidant activity.

3.8. Stability analysis after *in vitro* simulated gastrointestinal digestion

To investigate the protective effects of $W_1/O/W_2$ emulsions stabilized by native SPI or different complexes on *L. plantarum* A114, the viable count of *L. plantarum* A114 after *in vitro* simulated gastrointestinal digestion was detected by flow cytometer to evaluate the digestive resistance of *L. plantarum* A114 encapsulated by native SPI or different complexes (Fig. 11). The strain survival rate in the control group almost lost its activity after simulated gastrointestinal digestion (Qin et al., 2021). In addition, in the first 2 h of simulated gastric digestion, the viable count of encapsulated *L. plantarum* A114 decreased continuously. Specifically, the survival rates of strains encapsulated by the emulsions stabilized by native SPI or different complexes were $39.23 \pm 0.58\%$, $46.46 \pm 0.73\%$, $52.60 \pm 0.37\%$, and $64.22 \pm 0.89\%$ respectively, indicating that the emulsions could significantly increase the viable count of *L. plantarum* A114 after simulated gastric digestion (Frakolaki et al., 2020). Among them, the (EGCG-SPI)-EPS complex stabilized emulsion had a better protective effect on *L. plantarum* A114. In the (EGCG-SPI)-EPS complex, SPI exhibited strong resistance to gastric acid (Liu et al., 2023). Moreover, the interaction among SPI, EGCG and EPS also reduced the hydrolysis and erosion of gastric fluid (Sarkar et al., 2009). After further incubation in simulated intestinal fluid for 2 h, the survival rates of *L. plantarum* A114 showed similar trend with that after simulated gastric digestion. Where, (EGCG-SPI)-EPS complex stable emulsion provided the best protection compared to other groups. This might be due to the addition of EGCG and EPS formed a thick interface layer on the oil droplet surface and created a continuous network structure to prevent pancreatic enzymes from approaching the oil

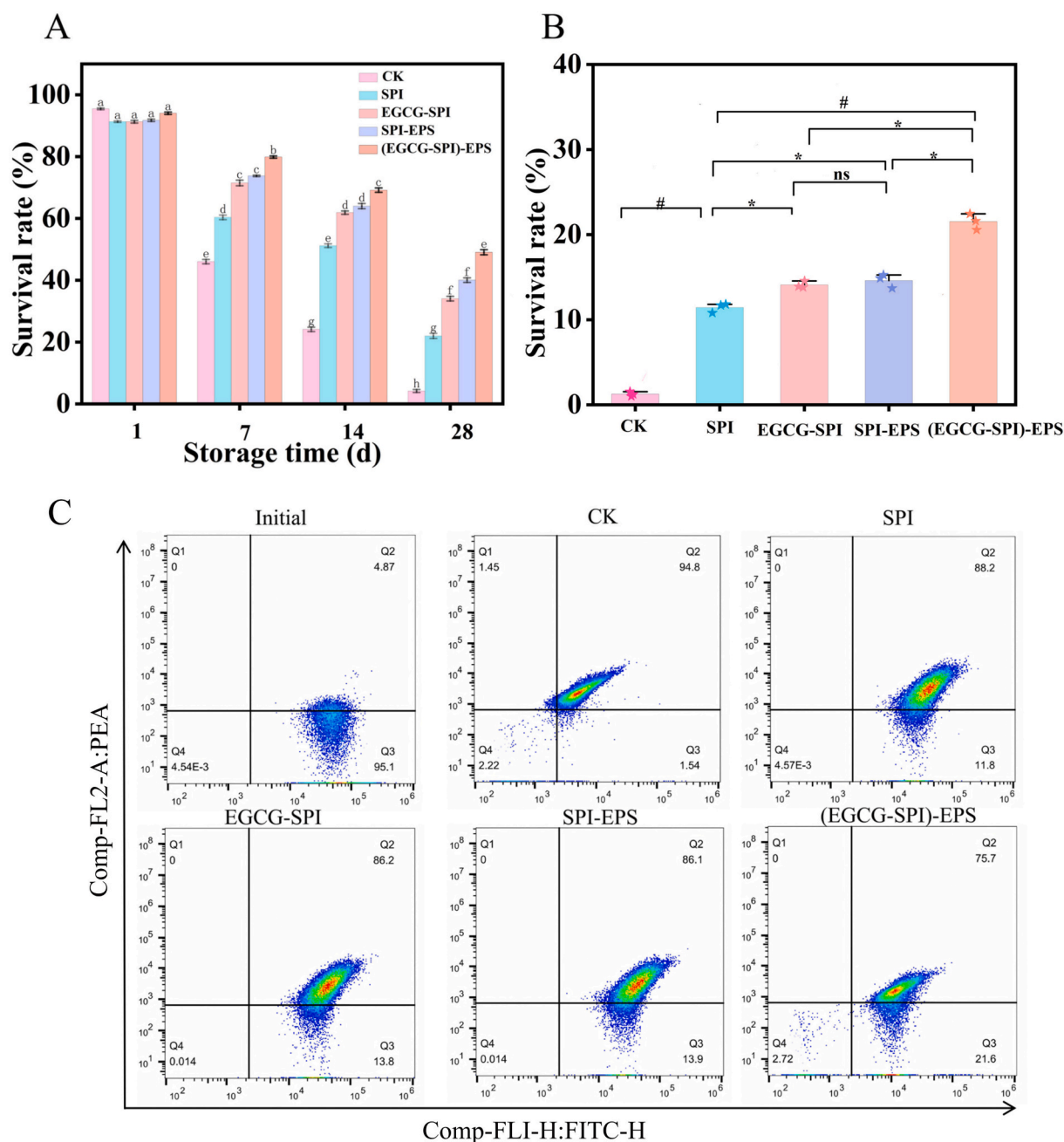


Fig. 10. The survival rates of *L. plantarum* A114 during storage (A) and after pasteurization (BC) were detected by flow cytometer. * indicates significant differences at $p < 0.05$, and # indicates significant differences at $p < 0.01$.

droplets and inhibit their decomposition (Qin et al., 2021). These results showed that (EGCG-SPI)-EPS complex could effectively enhance the resistance of *L. plantarum* A114 to the complex gastrointestinal environment. The survival rate surpassed the results reported by another study which mainly relied on strong interfacial stabilization and the physical barrier provided by medium chain triglyceride to provide physical protection efficiency for *L. reuteri* (Marefati et al., 2021). In contrast, the present study showcased the superior synergistic protective advantages of a multicomponent complex under harsh gastrointestinal conditions. Another study found that the survival rate of Bifidobacterium encapsulated in emulsion was significantly higher than that of the unencapsulated strain, which supported the current research results (Frakolaki et al., 2020). In summary, the protective effects under three typical stress conditions fully elucidated the synergistic protective mechanism of the $W_1/O/W_2$ emulsion stabilized by the (EGCG-SPI)-EPS

ternary complex on encapsulated *L. plantarum* A114, which was intuitively illustrated in Fig. 12.

4. Conclusion

This study investigated the stabilization of binary or ternary complexes to $W_1/O/W_2$ emulsions as well as the protective effect of different emulsions on *L. plantarum* A114. Compared with the emulsions stabilized by native SPI, EGCG-SPI, and SPI-EPS complexes, (EGCG-SPI)-EPS complex stabilized emulsion exhibited the best physical stability over a 28-day storage period. Specifically, it showed the smallest particle size change, the lowest degree of emulsion separation and excellent structural integrity. Furthermore, the covalent modification of EGCG enhanced the mechanical strength and antioxidant capacity of SPI interface film, while the introduction of EPS significantly improved the

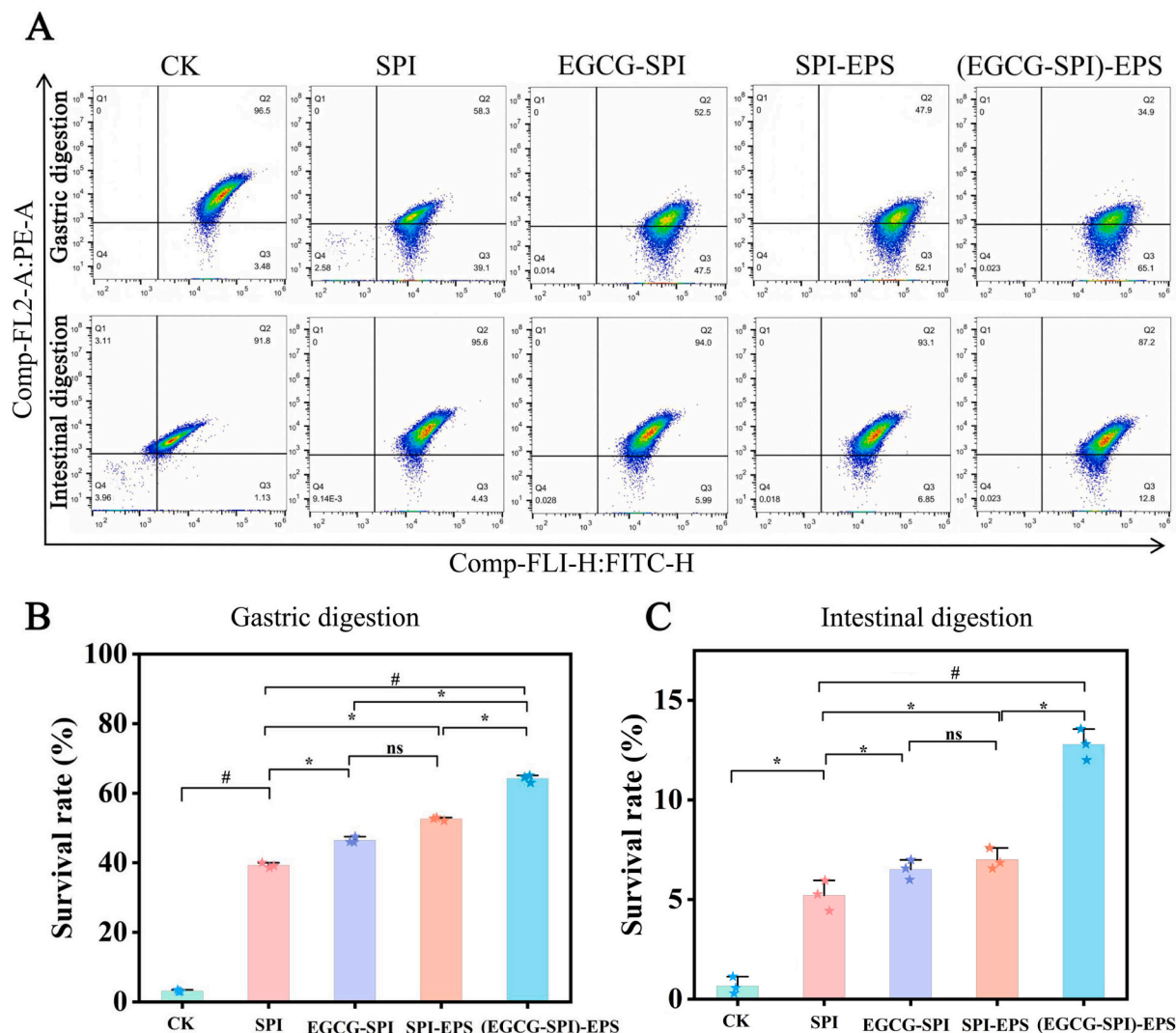


Fig. 11. The survival rate of *L. plantarum* A114 after *in vitro* simulated gastrointestinal digestion was detected by flow cytometer (A). B: The survival rate of *L. plantarum* A114 after *in vitro* simulated gastric digestion; C: The survival rate of *L. plantarum* A114 after *in vitro* simulated intestinal digestion. * indicates significant differences at $p < 0.05$, and # indicates significant differences at $p < 0.01$.

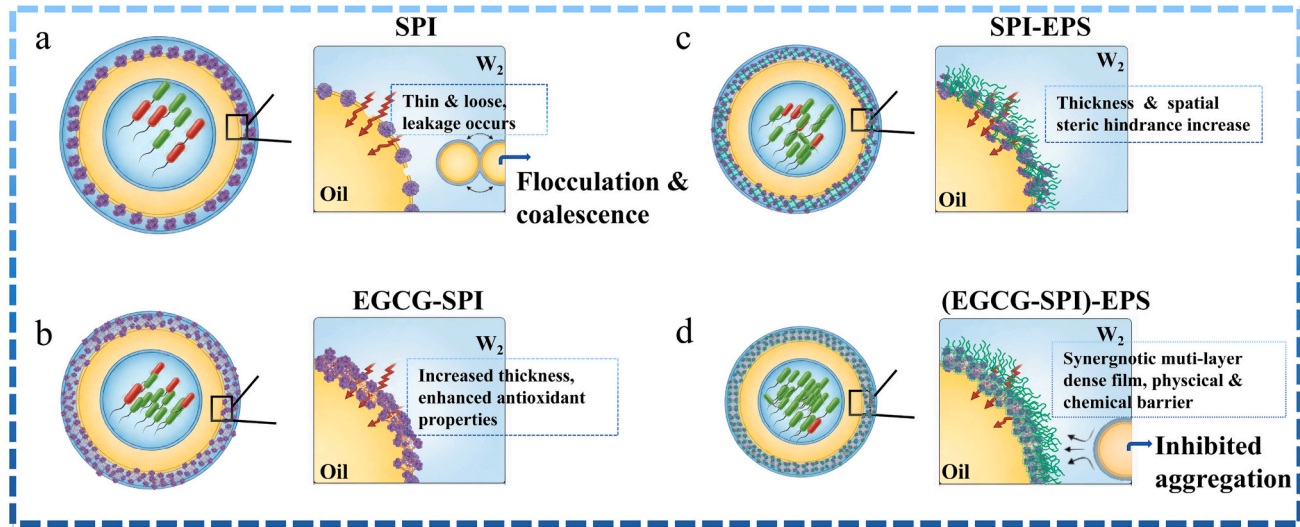


Fig. 12. Schematic illustration of the synergistic protective mechanism of (EGCG-SPI)-EPS ternary complex-stabilized $W_1/O/W_2$ emulsion on encapsulated *L. plantarum* A114 against storage, pasteurization and *in vitro* simulated gastrointestinal digestion stresses.

physical stability of the emulsion through spatial hindrance and continuous phase viscosity increase. Moreover, the ternary complex stabilized emulsion demonstrated outstanding protective ability during 28-day storage, pasteurization, and simulated gastrointestinal digestion processes. This proved its ability to effectively resist various stress factors such as heat, acid, enzymes, and oxidation. The (EGCG-SPI)-EPS composite stabilized emulsion, constructed through the synergistic formation of EGCG covalent modification and EPS interface assembly, possessed both excellent physical barrier function and biological activity protection capability, making it an ideal carrier system for targeted delivery and efficient protection of probiotics. This study provided theoretical basis and practical paths for developing high-efficiency probiotic encapsulation carriers based on the (EGCG-SPI)-EPS ternary complex, and had reference significance for the development of functional probiotic delivery systems.

CRediT authorship contribution statement

Cuicui Duan: Writing – review & editing, Funding acquisition, Formal analysis. **Cheng Wu:** Writing – original draft, Methodology, Data curation, Conceptualization. **Hui Liu:** Software, Conceptualization. **Siyang Sui:** Methodology, Investigation. **Hongru Wang:** Visualization, Methodology. **Lin Ma:** Supervision, Investigation. **Fumin Ma:** Visualization, Investigation. **Xiaolei Li:** Software, Investigation. **Dan Li:** Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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